Executive Insights Dashboard - Telecom Churn Dataset

"Predict behavior to retain customers. You can analyze all relevant customer data and develop focused customer retention programs."

Content

The Orange Telecom's Churn Dataset, which consists of cleaned customer activity data (features), along with a churn label specifying whether a customer canceled the subscription, will be used to develop predictive models. Two datasets are made available here: The churn-80 and churn-20 datasets can be downloaded.

The two sets are from the same batch, but have been split by an 80/20 ratio. As more data is often desirable for developing ML models, let's use the larger set (that is, churn-80) for training and cross-validation purposes, and the smaller set (that is, churn-20) for final testing and model performance evaluation.

Telecom Churn Datasets

Each row represents a customer; each column contains customer's attributes. The datasets have the following attributes or features:

State: string

Account length: integer

Area code: integer

International plan: string

Voice mail plan: string

Number vmail messages: integer

Total day minutes: double

Total day calls: integer

Total day charge: double

Total eve minutes: double

Total eve calls: integer

Total eve charge: double

Total night minutes: double

Total night calls: integer

Total night charge: double

Total intl minutes: double

Total intl calls: integer

Total intl charge: double

Customer service calls: integer

Churn: string

The "churn-bigml-20" dataset contains 667 rows (customers) and 20 columns (features).

The "Churn" column is the target to predict.

ANALYSIS GUIDE

1. Analysis Objectives

- **Objective 1:** Understanding the performance of customer service within the Orange Telecom churn (Identifying if customer service interaction is a predictor of churn).
- **Objective 2:** Analyzing Plan Subscriptions and Usage Patterns as Drivers for Customer Churn (Identifying high-risk segments based on their international/voice mail plans and billing).

2. Research Questions

For Objective 1 (Customer Service Performance):

1. **Correlation Analysis:** What is the relationship between the frequency of customer service calls and the probability of a customer churning?
2. **Threshold Identification:** Is there a specific number of service calls (e.g., 3 or more) that serves as a "tipping point" for customer dissatisfaction?
3. **Customer Behavior:** Do churned customers exhibit a higher average call frequency compared to loyal customers?
4. **Geographic Variation:** Are there specific states or area codes where customer service call rates are significantly higher, indicating regional service issues?

For Objective 2 (Plan & Usage Drivers):

1. **International Plan Impact:** How does the churn rate of customers with an International Plan compare to those without?
2. **Revenue vs. Churn:** Is there a significant difference in the total daily and international charges between churned and retained customers?
3. **Voice Mail Retention:** Does having a Voice Mail plan correlate with higher customer loyalty?
4. **Tenure Analysis:** Does the "Account Length" (tenure) of a customer influence their likelihood to churn, and are newer customers more at risk?

3. Tailored Goals for Solving Questions

- **Goal 1:** Identify the "Critical Call Threshold" to trigger automated retention offers or escalated support once a customer reaches a certain number of service calls.

- **Goal 2:** Evaluate regional service quality by mapping high-call states to infrastructure issues or localized competitor activity.
- **Goal 3:** Review the value proposition and pricing of the International Plan, as it may be driving customers away if perceived as too expensive or poor quality.
- **Goal 4:** Design "Loyalty Milestone" programs based on account length to reward long-term customers and reduce churn in newer segments.
- **Goal 5:** Optimize the Customer Service workflow to resolve issues in fewer than 3 calls.
- **Goal 6:** Create "Plan Bundles" that encourage Voice Mail adoption, as data suggests it acts as a retention anchor.
- **Goal 7:** Conduct a price sensitivity analysis by comparing total charges against competitor benchmarks.
- **Goal 8:** Segment customers into "Risk Tiers" (High, Medium, Low) based on a combination of service calls and usage volume.

4. Power BI Analysis Strategy

To analyze this dataset effectively in Power BI, follow these steps:

1. **Data Transformation (Power Query):**

   Data Typing: Ensure Churn and plan columns are converted to 'Yes/No' or 'True/False' format.

   Calculated Columns: Create a Total Charge column (Total Day Charge + Total Eve Charge + Total Night Charge + Total Intl Charge).

   Grouping: Create a "Call Frequency Bin" (e.g., 0-1 calls, 2-3 calls, 4+ calls) to simplify the Customer Service analysis. Use

2. **Data Modeling:**

   Since it is a single table, ensure measures are organized. Create a Measures table for Churn Rate, ARPU (Average Revenue Per User), and CS Call Average.

3. **Visualization Suggestions:**

   Churn Overview Dashboard: Use a **Gauge Chart** for the overall Churn Rate and **Donut Charts** for Churn by Plan type.

   Geospatial Analysis: Use a **Map Visual** using the State field to show Churn % by region.

Behavioral Trends: Use a **Decomposition Tree** to see how Churn is broken down by Customer service calls -> International Plan -> Total Charge.

Risk Analysis: A **Scatter Plot** showing Total Day Minutes vs. Total Day Charge, color-coded by Churn status.

5. Key Performance Indicators (KPIs) & MoM Suggestions

Key Performance Indicators:

- **Churn Rate (%):** The percentage of total customers who left Orange Telecom.
- **CS Sensitivity Index:** Average number of service calls per churned customer vs. retained customer.
- **ARPC (Average Revenue Per Churner):** The average revenue lost per customer who churns.
- **Plan Churn Ratio:** Comparison of churn rates between International Plan holders and Non-holders.

Suggestions for MoM (Month-over-Month) Performance:

The current dataset lacks a "Date" column but includes **Account Length**. To get a "Contact of the data" like MoM:

- **Proxy Timeline:** Treat Account Length as a proxy for time. Group customers by "Tenure Month" (Account Length / 30) to see if churn is higher in the first 3 months vs. the first year.
- **Simulate Snapshots:** In a real production environment, you would join this dataset with a "Billing Date" table. In Power BI, you can use **DAX Time Intelligence** functions (like SAMEPERIODLASTYEAR or DATEADD) once a date field is added.
- **Trend Analysis:** Create a line chart using "Tenure" as the X-axis to visualize the "Customer Lifecycle" and identify the specific month where churn spikes.

Create a measure table and measures using these steps

Step 1: Create the Placeholder Table

1. Go to the **Home** tab in the top ribbon.
2. Click **Enter Data**.

3. In the window that appears:

   Change the **Table Name** (at the bottom) to _Measures (The underscore keeps it at the very top of your list).

   Leave the column as "Column1" and don't enter any data.

   Click **Load**.

Step 2: Create the 3 Measures

Now, right-click your new _Measures table and select **New measure** for each of the following:

1. Churn Rate

This calculates the percentage of customers who have left.

Code syntax

```
Churn Rate =
DIVIDE(
   CALCULATE(COUNTROWS('Telecom Churn'), 'Telecom Churn'[Churn] = TRUE),
   COUNTROWS('Telecom Churn'),
   0
)
```

*(Format this as a **Percentage %** in the Measure Tools tab).*

2. ARPU (Average Revenue Per User)

This calculates the average total bill per customer. (Note: This assumes you have already created a Total Charge column, or you can use the sum of all charges as shown below).

Code syntax

```
ARPU =
AVERAGE ('Telecom Churn' [Total charge])
```

*(Format this as **Currency**).*

3. CS Call Average

This helps track the average number of times a customer contacts support.

Code syntax

CS Call Average = AVERAGE (“Telecom Churn’ [Customer service calls])

*(Format this as a **Decimal Number**).*

Step 3: The "Magic" Cleanup

To make it a true Measure Table (and get the special icon):

1. In the **Fields** pane, click on the three dots ... next to the empty **Column1** in your _Measures table.
2. Select **Hide**.
3. **Collapse and Re-expand** the Fields pane (or click the little arrow next to the table name).
4. The icon for the _Measures table will change from a table icon to a **Calculator icon**, and it will move to the very top of your list.

Now to create the "Previous Month" Measure

To calculate MoM, Power BI needs to look at the "Current Month" in the dataset and then find the value for "Current Month minus 1."

1. Right-click the **Measures** table and select **New Measure**.
2. Enter this formula:

Code syntax:

```
Churn Rate (Prev Month) =

VAR CurrentMonth = SELECTEDVALUE ('Telecom Churn'[Tenure Month])

RETURN

CALCULATE (

   [Churn Rate],

   ALL ('Telecom Churn'[Tenure Month]),

   // Remove the current month filter

   'Telecom Churn'[Tenure Month] = CurrentMonth - 1

)
```

Step 3: Create the MoM % Change Measure

Now that you have the current rate and the previous rate, calculate the percentage difference.

1. Create another **New Measure**:

Code syntax

```
Churn MoM % Change =

VAR CurrentRate = [Churn Rate]

VAR PrevRate = [Churn Rate (Prev Month)]

RETURN

IF(

   ISBLANK(PrevRate),

   BLANK(),

   DIVIDE(CurrentRate - PrevRate, PrevRate)

)
```

2. Format this as a **Percentage %**. Use **Red** for positive (increasing churn) and **Green** for negative (decreasing churn) using Conditional Formatting.

Because this is a "Static Snapshot" dataset (one row per customer), the MoM calculation here represents **how churn risk evolves as a customer gets older**, rather than how the company performed in January vs. February.